You can run Keycloak locally as a standalone Java application or a container. For production, there are a few solutions for running Keycloak on Kubernetes. Keycloak also needs a relational database for persistence. It comes with an embedded H2 database, but you'll want to replace it with an external one in production.

For Polar Bookshop, we'll run Keycloak locally as a Docker container, relying on the embedded H2 database. In production we'll use PostgreSQL. This might seem to contradict the environment parity principle, but since it's a third-party application, it's not your responsibility to test its interaction with a data source.

This section will guide you step-by-step through the Keycloak configuration for the Polar Bookshop use case. First, open your `polar-deployment` repository. Then define a new `polar-keycloak` container in docker/docker-compose.yml.

**Listing 11.3 Defining a Keycloak container in Docker Compose**

```
version: "3.8"
services:
  ...

  polar-keycloak:                              ❶
    image: quay.io/keycloak/keycloak:19.0
    container_name: "polar-keycloak"
    command: start-dev                         ❷
    environment:                               ❸
      - KEYCLOAK_ADMIN=user
      - KEYCLOAK_ADMIN_PASSWORD=password
    ports:
      - 8080:8080
```

❶ The section describing the Keycloak container

❷ Starts Keycloak in development mode (using an embedded database)

❸ Defines admin credentials as environment variables